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Education

York University

Bachelor of Science in Computer Science (Honours)

January 2020 – December 2024

Toronto, Canada

Work Experience

Privex

Software Engineer

October 2023 – January 2024

London, United Kingdom

- Developed an AI customer service agent using RAG techniques on a self-hosted LLM to handle common customer inquiries related to technical issues, products, and services.
- Developed REST APIs in Flask for querying and training the LLM, integrated with Discord APIs to allow customers to query the AI agent and staff to train it with new data on demand.
- Utilized vector databases and MySQL for training data management. Developed scripts to remove existing data from the database and to add new data from the website's documentation.

CogniPet

Software Engineer

April 2019 – January 2020

Zürich, Switzerland

- Developed the Pet Breed Detection system by training a deep learning model using the Xception neural network in Keras to recognize 120 breeds of dogs with 93% accuracy.
- Implemented REST APIs in Flask for model inference and integrated them with CogniPet's Android and iOS apps using Amazon Web Services (AWS).
- Developed proof of concept for a facial recognition system in pet identification by collecting data from dogs and using video-based facial recognition techniques.

Projects

Jix Interpreter | *C Interpreter*

- Developed an interpreter for a dynamically typed programming language with support for basic functional programming.
- Implemented fault-tolerant parsing and error handling, dynamic types, arrays, functions with nested blocks and recursion, built-in functions, conditional statements, loops, and control flow statements.
- Added functional programming support, including first-class functions, higher-order functions, and function pointers.
- Tested the interpreter using automated tests written in Jix and integrated with CI/CD pipelines.

Prism Coverage | *Java Code Coverage Engine*

- Developed a code coverage engine supporting control flow (statement, decision, condition, path) and data-flow (All-Defs, All-DU-Paths, All-P-Uses) coverage criteria for a programming language.
- Implemented coverage for built-in and user-defined types, conditional statements, functions with nested blocks, and loops.
- Generated visual HTML reports with color-coded coverage details and summaries for control-flow and data-flow coverage.

Cap Containers | *C Containers Library*

- Developed a highly generic, header-only containers library in C.
- Supported sets, maps, vectors, hash-tables (open and closed addressing), queues (circular, dynamic, fixed, and priority), stack, lists, and an arena memory allocator.
- Included thread-safe variants with cross-platform support (Windows, Linux, and macOS).
- Tested using automated tests and documented with Doxygen.

SOCKS5 Handler | *C++ SOCKS5 Library*

- Developed a library implementing RFC 1928 and RFC 1929 for a SOCKS5 client with username/password authentication and remote DNS resolution; achieved 30 stars on GitHub.

Technical Skills

Languages: Python, C++, C, Java, Go, JavaScript, SQL, Perl, Bash, PHP, HTML/CSS, L^AT_EX

Technologies: Linux/Unix, MySQL/PostgreSQL, Django/Flask, JWTs, CI/CD pipelines, Neo4j

Developer Tools: Git, Docker, LXC/LXD, GDB/LLDB, Amazon Web Services